Math 252, Second Examination-Review

Directions: Respond to each question in the space provided, under each question. You may use the back of the page if necessary. All calculations and mathematical work must be shown and justified, including citation of well known Theorems or properties that have been used for solving the problem. Only a TI-30X (type) scientific calculator may be used on this exam. This is a closed note, closed book exam. No notes, laptops, smartwatches, mobile devices, or any other unapproved electronic device may be used while taking this exam.

1. (a) Evaluate the integral: $\int xe^{-3x}\,dx$.

   (b) Using your result from part (a), evaluate $\int_0^\infty xe^{-3x}\,dx$ to determine whether it converges or diverges.

2. (a) Evaluate the integral: $\int \frac{\ln x}{x^3}\,dx$.

   (b) Using your result from part (a), evaluate $\int_0^1 \frac{\ln x}{x^3}\,dx$ to determine whether it converges or diverges.

3. Evaluate $\int \sin^3 x \cos^3 x\,dx$.

4. Evaluate $\int \tan^3 x \sec^3 x\,dx$. .

5. Evaluate $\int \frac{dx}{x^2\sqrt{x^2+4}}\,dx$.

6. Evaluate $\int \frac{\sqrt{x^2-9}}{x}\,dx$.

7. Evaluate $\int \frac{2x+6}{x^2+2x}\,dx$.

8. Evaluate $\int \frac{dx}{x(x-3)^2}$

9. Evaluate $\lim_{x\to 0^+} x^{2x}$

10. Evaluate $\lim_{x\to 0^+} x\ln x$